

Stage:	Class(es):	Description	Formula:	Location:
Setup	BatteryRun & Batt	The PASA availability is extracted from the Last Acknowledged Bid for each battery and attached to each battery object.		line 3 man line 3 batt.
Extraction	BatteryInputBase	For all batteries, the most recent SoC and capability values are extracted, as well as the closest SoC reading to the beginning of the current dispatch interval.		line 3 input
Extraction	BattRivDarllInputBase	For RESS2 and DPNTBESS, the throughput over the previous 7.5hrs is also extracted as follows:	Throughput = (max(cumenergyexported) - min(cumenergyexported) + max(cumenergyimported) - min(cumenergyimported))*100	line 3 input
Extraction	AvailInput	Query the availability data from the UI for all batteries, includes current trading day and D+1 trading day, if available. Extract the first interval with LOAD availability form the AvailInput and		line 3 input line 3 input

		assign to the Batt objects.		
Validation	BatteryInputBase	Checks the age of the SoC and capability readings. If the reading is more than 10 minutes before init_dt, flag the battery to be bid unavailable.		line input
		If there is no data returned, fail the validation and exclude the battery from the optimisation.		line input
Transform	BatteryInputBase	Calculate the minutes between the latest reading and init_dt, then estimate the SoC at init_dt:	$\text{init_soc} = \text{last_reading} + (\text{LOAD} - \text{GEN} - \text{REG_R_UTIL} + \text{REG_L_UTIL}) * \text{min_remaining} / 60$	line input
		Estimate the min and max projected SoC values at init_dt as the latest reading +/- the maximum load/generation for the minutes between that reading and the init_dt.	$\begin{aligned} \text{max_init_soc} &= (\text{last_reading} + \text{mins_prior_to_curr_di} * \text{bess.serv_max}['\text{LOAD}'] / 60 + \text{mins_left_in_curr_di} * \text{disp}['\text{LOADDUID_TOTAL}'] / 60) \\ \text{min_init_soc} &= (\text{last_reading} - \text{mins_prior_to_curr_di} * \text{bess.serv_max}['\text{GEN}'] / 60 - \text{mins_left_in_curr_di} * \text{disp}['\text{GENDUID_TOTAL}'] / 60) \end{aligned}$	line input

		Extract the latest Capability values for both Charge and Discharge and assign to battery objects.		line 1 input
Transform	BattRivDarInInputBase	Extract the throughput values and estimate the maximum throughput as the actual throughput plus the maximum throughput between the reading and init_dt, plus a 5 MWh buffer.	adjusted_throughput_mwh = (throughput_mwh + mins_since_reading * nameplate_capacity_mw / 60 + buffer_mwh)	line 1 input
		Extract the most recent fullpackenergy value.		line 1 input
Transform	AvailInput	Cap LOAD and GEN using the PASA availability from the last ack bid.		line 2 input
		If the battery data is older than 10 min, set the availability for the next <i>n</i> intervals to 0 MW.		line 2 input
		Else: If the minimum potential SoC < min	Availability = 12 * (min_proj_soc-params['MIN_SOC_MWH'])	line 2 input

		SoC, limit GEN DUID availability in the next interval to: Then set availability in the following <i>n</i> intervals to 0 MW.		
		If the max potential SoC > max SoC, limit LOAD DUID availability in the next interval to: Then set availability in the following <i>n</i> intervals to 0 MW.	Availability = 12 * (params['MAX_SOC_MWH'] - max_proj_soc)	line 2 input
		If there is an LPA that may cause unfavourable discharge or charging, set availability to 0 MW for the next <i>n</i> intervals.		line 2 input
		If DP or RESS are able to violate the throughput limits in the next 30 minutes, set availability to 0 MW for the next 30 minutes.		line 2 input
		Limit GANNBG1 availability using the GSF forecast with a buffer added.	Avail = 50 - (GSF_forecast + buffer) MW	line 2 input

		Limit GANNBL1 to 5 MW if peak hours, else leave it unchanged.		line 2 input
		If battery availability in the next <i>n</i> intervals is greater than the most recent battery capability value, limit the next <i>n</i> intervals to the capability for each DUID.		line 2 input
Transform	AvailInput	Set a column in the AVAIL data for the MAX_SOC_MWH based on the MAX_SOC_MWH param.		
		Restrict the first <i>n</i> intervals using the fullpackenergy value for DP and Riverina only.	$\text{MAX_SOC_MWH}(t=0, t=n) = \text{MIN}(\text{fullpackenergy}, \text{MAX_SOC_MWh}(\text{UI}))$	line 2 input
		Set the HIGH and LOW SoC thresholds based on the % values from the UI.		line 2 input
Transform	AvailInput & Batt	To determine if a battery will cause an infeasible error in the Planner, the SoC at the first interval with LOAD	$\text{projected_soc} = \text{init_soc} - \text{aux_mw} * \text{hours_until_load_avail}$	line 2 batt.

		availability is calculated as:		
		A second infeasible check is performed to ensure the battery has enough SoC and LOAD availability in the first interval to stay above the minimum limit.	$\text{soc_1_mwh} = \text{init_soc} + (\text{initial_load_availability} * (1 - \text{char_loss}) - \text{aux_mw}) / 12$	line batt.
		If battery is infeasible, set availability for all services and all intervals to 0 MW for that battery.		line input
Transform	AvailInput	Flag all of the above AVAIL restrictions or modifications as absolute variables in the silver data.		

Executive Summary

Situation

- EA has partnered with Wartsila to setup a battery energy storage site (BESS) located in the same area as the Jeeralang Gas energy generation site.
- The Wooreen BESS will be registered with AEMO as a new EA Asset to enable energy trading and dispatch.
- The new site will use the standard applications used in existing EA OT sites:
 - Gallagher – Access Control
 - ePAS – Work Permit Management
 - Milestone – Security Camera Management

- Maximo – Asset Management
- Objective – Asset Document Library
- PI – Centralised OPC Data Repository
- Wartsila will provide the battery storage and controller application.
- Yurika will be responsible with the integration and network setup.

Complications

- The current data feed from PI to the EA Battery Bidding system has issues around accuracy, frequency and availability.
- The license model for the ePAS application has changed from perpetual to subscription-based which made it more expensive.
- Remote access to HMI Client in the OT network by EA energy traders (IT network) is required to get “missing” Wooreen OPC data in the SCADA and WDS applications.
- The PI Data Integration architecture in Wooreen must address the current data issues raised by the EA Battery Bidding team.

Questions

- What PI Data integration method between the Wooreen Battery Controllers and the Battery Bidding application will ensure that issues in Riverina will not be encountered in Wooreen?
- Does the Maximo Work Permit feature contains all the functionalities required for it to be a viable replacement for the ePAS application?
- How will Wooreen onsite staff get remote access to the Maximo, Objective, and ePAS applications that are in the EA IT network?

Resolution

- The existing Jeeralang MPLS WAN network and OT Firewall will be leveraged for network connectivity between Wooreen OT and EA IT. A dedicated fibre connection will be used to connect the Wooreen OT applications to the Jeeralang network.
- For security reasons, a uni-directional data traffic between Wooreen OT and EA IT will be enforced through the use of data diodes.
- Unlike in Riverina, the Wooreen battery data will be streamed to the PI data server in EA IT via the DA (Data Access) protocol instead of the UA (Unified

Architecture) protocol. This approach takes advantage of the PI DA Interface which is more stable and reliable than the PI UA Connector used in Riverina.

- There will be two instances of the PI Integration servers in Wooreen to ensure high availability of OPC data streams to the PI Data server.
- Wartsila will also implement a high availability architecture for their GEMS Power Plant Controllers to ensure stable integration with the battery storage units, AEMO (energy dispatch), and EA Trading platform (battery data).
- Remote access to EA IT applications (Maximo, Objective, and ePAS) by Wooreen onsite staff will be provided via the Azure Application Proxy.
- Security Cameras and audio devices installed in Wooreen site will be integrated with the existing Milestone server in Jeeralang or to a new Milestone server in Wooreen site. This will depend on the current capability of the existing Milestone server Jeeralang to handle the increased load.
- The Maximo Work Permit module can be used in a pilot program for Wooreen to determine if it can be a viable alternative to ePAS for EA operation sites. This module is already included in the current Maximo license and the small size of the onsite staff makes Wooreen a good candidate for this pilot program.
- HMI is part of the deliverables from Wartsila and their current proposal for remote access to the controllers via the GEMS Fleet Director is not approved by Security. A different approach for remote access to the HMI Client will have to be explored with the vendor.

Conceptual View

As-Is

Not applicable as this is a new battery energy storage site.

To-Be

Wooreen Battery Energy Storage Site Conceptual Domain Architecture

GEMS Power Plant Controller (PPC) Architecture

EA has engaged Wartsila to provide the batteries, inverters, and the power plant control and energy management system at Wooreen. Wartsila will install onsite their GEMS Power Plant Controller (PPC) application and has also proposed their cloud-hosted GEMS Fleet Director (FD) to enable plant operations to be done remotely. As a cyber security requirement, remote access to operations site is not allowed therefore the GEMS FD will not be used in Wooreen (for example if the Wartsila cloud app is

compromised, then a bad actor can get complete control of the Wooreen site). A uni-directional traffic for outbound data from GEMS PPC to the EA corporate network will be enforced through data diodes.

The GEMS PPC application has a built-in OPC UA Server that will be used to integrate with the EA PI Central database. Two GEMS PPC instances are deployed in the GEMS PPC Stratus server in a hot standby with failover arrangement for high availability and dual connectivity to EA IT. The server will sit in a security zone separated by an internal firewall (TBC) from the batteries and inverters controlled by the GEMS PPC application. GEMS PPC has an “Automated Operation” mode whereby it will use the build-in Rule Engine to perform power operations based on detected conditions such as market instructions, energy price, equipment or plant conditions. Manual operations by onsite staff can be done through the GEMS PPC web-based UI.

Yurika is the OT integration partner and will be responsible for meeting the network and connectivity requirements.

Only the GEMS Power Plant Controller servers will be located in the Wooreen Control Room. The rest of the OT applications like the PI Data Interface servers will be located in the Wooreen O&M Room that is connected to the Control Room via dedicated fibre cable.

Inbound and outbound data traffic between the battery/inverters and the two PPC instances in the GEMS server will be managed through network switches as shown in the diagram above. Wartsila’s standard network design includes firewalls between the network switches and each battery/inverter device which may be a bit of an overkill and unnecessary for Wooreen since the entire stack is already behind an OT firewall.

Inbound data initiated by the GEMS power plant controller will be routed to the batteries/inverters via the network switches AF21 and AF31. Outbound data from the batteries/inverters to the GEMS power plant controller will be via AF32 and AF22. The network switches AF21 and AF22 will be configured to provide automatic failover if one of the power plant controllers is not available/unhealthy.

Outbound data from the UA Server in the GEMS PPC to the PI Interface servers will be via the AF11 and AF12 routers which have built-in firewalls.

ChemAlert

ChemAlert by Risk Management Technologies (RMT) is a chemical management SaaS application used to ensure that the correct controls and protections are in place in Wooreen for chemicals stored onsite. Features include:

- Identify, record, and search for on-site chemicals through barcodes

- Access to the latest high-quality safety data sheets
- Find and print the latest GHS compliant labels
- Set up Stock Manifests, Reports – Hazardous/Dangerous Chemicals, Placarding and Storage Incompatibilities.
- Document risk assessments with ease and transparency, ensuring appropriate controls are in place.

ChemAlert is used on all sites that EA own and operate.

Asset Management using Maximo

Onsite staff will have remote access to the Maximo application through the Azure Application Proxy service, which is possible since Maximo has a web client accessible from an internet browser (single endpoint). The Wooreen Control Room will be connected to the internet and EA via the existing Jeeralang network. The Azure Virtual Desktop is an option but not recommended for the following reasons:

- It would require a Remote Desktop Connector installed on the staff's device
- It consumes Virtual Desktop resources that can come with a significant cost

Jeeralang support staff already uses Maximo and the Site Security Group (AD) for the Jeeralang staff can be used as a template for the Wooreen staff and provide the following security controls:

- Services/Support Maximo User Access Form (which allows access to multiple sites for System and Support users)
- Individual Site Maximo User Access forms can allow access to the one site there are designated for. Note: Access form for Wooreen staff cannot be used to access Jeeralang.

Milestone System Architecture

The security cameras and video management console in the Wooreen Control Room will be integrated with the Milestone video management system (VMS) installed in the Jeeralang Control Room, which is made up of the following servers:

Server Type	Location	Description

Management Server	Onsite - Jeeralang	The central component of the Milestone video management system that can be in the same machine as the management server or in a separate machine. It manages the surveillance security rules.
Recording Server	Onsite - Jeeralang	This is responsible for all communication, recording, and event handling. It handles the encryption of data streams to the clients and services as well as the event database. Database optimised for recording and storing audio and video data.
Event Server	Onsite - Jeeralang	This handles tasks related to system events and alarms and can also handle the event database.
SQL Server	Onsite - Jeeralang	This manages the central database used by the Management and Recording Servers.

The current network bandwidth allocated to the video management system needs to be assessed to ensure that it can handle the additional video and audio data streams coming from the Wooreen Control Room. The Milestone VMS also has a Mobile Device Server that can be used in the future to enable security staff to access recorded or real-time video and audio feeds from authorised mobile devices.

[Talk to Tim and Brenton for details of the current Milestone infrastructure in Jeeralang]

ePAS Work Permit Management

The ePAS application from Prometheus will be used to manage the work permits, isolations, risk assessments, and safety requirements in the Wooreen site. This application is currently installed in EA Azure Cloud and will be accessed remotely by onsite staff via the Azure Application Proxy service. Due to the small size of onsite staff at Wooreen, ePAS will be used as a standalone application and does not have to be integrated with other applications like Gallagher.

[ePAS data model - Jeeralang, Wooreen, Newport as separate and independent sites]

The license model for ePAS is changing from a perpetual license to a subscription model based on user count which will significantly increase the license cost. For this project, the Maximo Permit Module will be looked into as a possible alternative. This module is already covered in the current Maximo license and provides the following set of features/functions:

- Managing Permits to Work that allows for different types of work permits, identification of the risks and hazards associated with the work permit, comprehensive review and approval process for the work permit before qualified

personnel are assigned to the tasks in the work permit, a “Toolbox Talk” for communicating the safe working procedures associated to the tasks, facility for the staff assigned to perform the task to update the work permit based on the actions performed (i.e., completing permits to work).

- Eliminating hazards via lockout and tagout plans that describe how to safely remove and replace work assets.
- Create a bypass operation to allow for work that requires the bypass of a safety control system.
- Create, review, and approve certification records that will be used to verify the qualification of onsite personnel to perform their assigned tasks.
- Ensure regulatory compliance with audits and surveys.

The HSE component of Maximo is installed and available but there are currently no AD Groups defined to enable access and usage of this component. HSE contains the Permit Module. Also note that changing to Maximo from ePAS will increase scope of the project as it will require a new set of work instructions and user training.

Gallagher System Architecture

Gallagher is the security and access control system for sites that are owned/operated by EA. Gallagher controllers will be installed onsite to manage access to doors and alarms in the Wooreen Control Room. These devices will be integrated with the Gallagher Command Centre (GCC) application in EA Azure Cloud. These devices will still work when network connectivity to the GCC is not available but device configuration updates from GCC will be put on hold until connectivity is restored. Additional license may be required for the new controllers installed in Wooreen.

The current integration between Gallagher Command Centre and SAP Success Factors will not be leveraged in Wooreen due to the small number of onsite staff in this site. As an alternative, site administration data required for each device configuration will be entered manually in the GCC application. This will be done remotely via the Azure Application Proxy service.

OT to PI Data Integration Architecture

The integration architecture for battery data into PI will follow the pattern for energy sites that EA owns or co-owns. This pattern enforces a uni-directional flow of data from OT network to the IT network via a Data Diode which provides another layer of

protection of the OT systems from unauthorised 3rd party intrusion via public internet. This pattern also has the PI Interface deployed in the OT network which takes advantage of the data buffering feature of the interface that prevents data loss in the event of a connection issue between the OT and IT networks. The third feature of this pattern is the conversion of OPC data from the Unified Architecture (UA) to classic Data Access protocol. The collection and sending of OPC UA data to PI Central requires the use of the PI UA Connector which is not as stable and reliable as the PI DA Interface. OPC UA protocol is still fairly new and is not yet as widely supported as the older OPC DA protocol.

The Unified Architecture (UA) protocol is recommended over Data Access (DA) protocol due to the following advantages it has over DA: 1) UA is OS agnostic whereas DA is a Windows-specific product; 2) UA allows for more flexible data types; 3) UA is not tied to the COM/DCOM transport method unlike DA - i.e. it also supports TCP/IP and HTTPS. UA data integration with PI is via a PI Connector whilst DA integration is via PI Interface. PI Connector is built on a new integration architecture and is not as stable as a PI Interface therefore the OPC DA protocol will be used to ensure the reliability of data feeds via PI from the OT Battery sites to the IT Battery Bidding platform.

The Trading team has expressed the need to get data feeds directly from OPC Servers instead of getting them from PI Central. The reasons for this are as follows:

Issue	Possible Mitigation
Issue	Possible Mitigation
Stability of the PI UA Connector (OPC Server to PI Server) and the PI Integrator (PI Server to Battery Bidding Server)	The PI UA Connector can be replaced with the PI DA Interface (Data Access) protocol to DA (Direct Access) as it is based on old but reliable protocol to DA protocol before the change to UA Gateway and EA uses the Unified Architecture.
OPC Data is streamed from PI to the Battery Bidding platform every 5 seconds with a consolidated historical data collection sent every 5 minutes.	Data points are sent only when the value changes and are available for Battery Bidding. Consolidated historical data is sent every 5 minutes. Each tag can be configured with an alarm system. Real-time streaming is through the PI In-Memory Kafka. A possible alternative to PI In-Memory is PI In-Memory.

	PI has a data compression setting d tag data is only stored when the valu data on a given frequency/interval re
Process of merging or combining data points from multiple OPC servers (HA architecture) to a single feed to Battery Bidding is not simple	
Loss of comms (AEMO to Battery Controller) should trigger a failover from the primary controller to the stand-by controller.	HA architecture of the PPC servers HA architecture of PI servers (applic
Validating the accuracy of the PI Data can be difficult	Each PI Tag value is stored with a tim value whilst the quality flag can be u connectivity problem with the data s data point.

Battery Bidding Functional Requirements

ID	Functional Requirement
FR7.6.01	The Battery Bidding solution must be able to access the Riverina and Darlington Point batt Critical Data required: • State of Charge (MWh or kWh) updated at least once every 5 seconds • State of charge % updated at least once every 5 seconds • Cumulative discharge (MWh or kWh) updated at least once every 1 minute • Current Maximum capacity (MWh or kWh) updated at least once every 5 seconds • Realtime Frequency data (Hz) at a minimum of 1 second resolution Data required to manage any specific contractual requirements is also critical. A separate timestamp of the reading for each of the above fields is required. Note: All other battery site data needs to be available to the Battery Bidding solution.

	Ultimately the Battery Site data needs to be available in Databricks Delta Tables.
FR7.6.02	The Battery Bidding solution must receive the site data information in near real-time, less request..
FR7.6.03	Separate Service Accounts for Batteries Non-Prod and Prod environments (if applicable).
FR7.6.04	The Battery Bidding solution must be able to retrieve the last acknowledged bid set for all and Darlington Point battery DUIDs.
FR7.6.05	The Battery Bidding solution must be able to submit bids for the Riverina and Darlington Point batteries.
FR7.6.06	Battery Bidding solution must be able to query Infoserver data using the Riverina and Darlington Point battery DUIDs.
FR7.6.07	The solution will ensure Battery site data availability of 99.5% (Gold) for battery bidding system.
FR7.6.08	The Battery Bidding solution must have the ability to retrieve the latest record for the following: • State of Charge (MWh or kWh) • State of charge (%) • Cumulative discharge (MWh or kWh) • Current Maximum capacity (MWh or kWh)
FR7.6.09	The Battery Bidding solution must have the ability to retrieve the frequency data every 10 seconds for a specified period.